

UNIVERSIDAD TECNOLÓGICA NACIONAL

FACULTAD REGIONAL BUENOS AIRES

Segundo Parcial de Mecánica de los Fluidos

Alumno:

Curso:

Legajo:

Fecha: 10/11/08

TEMA 1

- 1) Para la instalación de la figura determinar la bomba más adecuada de acuerdo a las curvas indicadas. Determinar además la potencia de la misma.

Datos:

$$Q = 8 \text{ m}^3/\text{hr}$$

$$D_1 = 0.04 \text{ m}$$

$$D_2 = 0.05 \text{ m}$$

$$K_{\text{codo}} = 14 f$$

$$K_{\text{válv}} = 100 f$$

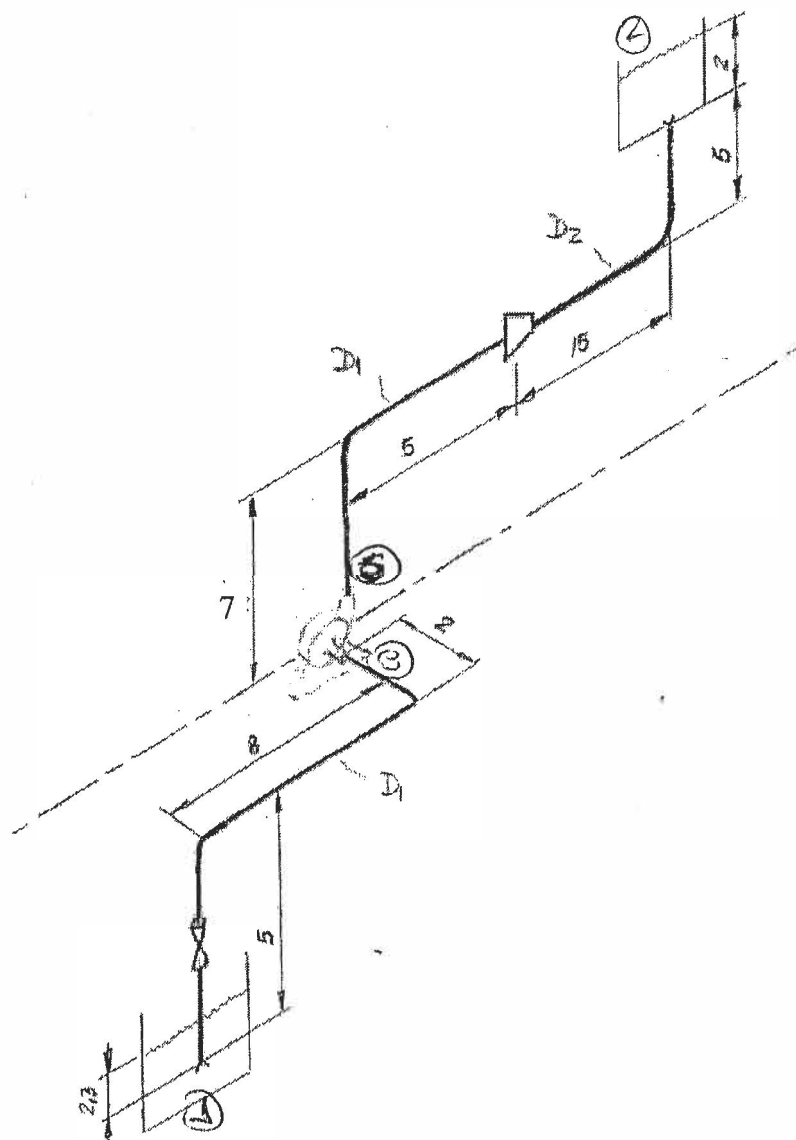
$$K_{\text{red}} = 300 f \text{ (aguas arriba)}$$

$$K_{\text{entrada}} = K_{\text{salida}} = 1$$

$$\epsilon = 25 \mu\text{m}$$

$$\rho = 1000 \text{ Kg/m}^3$$

$$\mu = 1 \times 10^{-3} \text{ Pa s}$$



TRIO

$$Q = 8 \text{ m}^3/\text{h} = 2,22 \text{ m}^3/\text{s}$$

$$\frac{P_e = 4,91}{S_g}$$

$$27,4398$$

$$Re = \frac{\rho V D}{\mu} = \frac{1000 \cdot 2,22 \cdot 0,04}{1 \times 10^{-3}} = 8,8 \times 10^4$$

$$\frac{\Sigma}{d_1} = 0,000225$$

$$V_2 = 1,4208$$

$$t_1 = 0,0175 \quad t_2 = 0,021$$

$$R = \frac{3 \cdot 2,22 \cdot 0,05}{1 \times 10^{-3}} = 1,11 \times 10^5$$

$$\frac{\Sigma}{d_2} = 0,0005$$

$$t_2 = 0,0192$$

$$t_{t_2} = 0,018$$

$$0,6 \text{ m}$$

$$\frac{V_s = 21,63}{S_g}$$

$$t_{\text{bambas}} = 21,63 - 4,91 = 16,72$$

$$\text{AmpA} =$$

plug.
Bojo

1)

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + \Delta h$$

$$\Delta h_{\text{cond}} = \frac{V_2^2}{2g} \cdot 14 \text{ ft}$$

$$\Delta h_{\text{cond}} = \frac{V_2^2}{2g} \cdot \frac{154}{0.049} \text{ ft}$$

$$\Delta h_{\text{val}} = \frac{V_2^2}{2g} \cdot 100 \text{ ft}$$

$$\Delta h_{\text{ent}} = \frac{V_2^2}{2} \cdot 1$$

$$\frac{101325}{1000 \cdot 9.8} = 2.7 + \frac{V_2^2}{2g} (1 + 1 + 475 \text{ ft} + 28 \text{ ft}) + \frac{P_2}{\rho g}$$

$$\frac{P_3}{\rho g} + \frac{V_3^2}{2g} + z_3 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + \Delta h$$

$$\Delta h_{\text{cond} 1} = \frac{V_3^2}{2g} \cdot \frac{12 \text{ ft}}{0.049}$$

$$\Delta h_{\text{cond}} = \frac{V_3^2}{2g} \cdot 14 \text{ ft}$$

$$\Delta h_{\text{val}} = \frac{V_3^2}{2g} \cdot 300 \text{ ft}$$

$$\frac{P_3}{\rho g} + \frac{V_3^2}{2g} = 10.33 + \frac{V_2^2}{2g} (468.75 \text{ ft}_1 + 21.875 \text{ ft}_1 + 468.75 \text{ ft}_3 + 400 \text{ ft}_2 + 14 \text{ ft}_2)$$

$$P_3 = 10.33 + \frac{V_2^2}{2g} (-0.3625 + 987.5 \text{ ft}_1 + 21.875 \text{ ft}_2 + 400 \text{ ft}_2 + 14 \text{ ft}_2) \cdot 1$$

$$V_1 = V_2 \frac{A_2}{A_1}$$

$$V_3 = V_2 \frac{A_2}{A_1}$$

$$V_3 = V_2 \frac{A_2}{A_1}$$

$$V_3 = V_2 \cdot 1.5625$$

$$V_2 = V_3 \frac{A_1}{A_2}$$

λ	ν	t_1	t_{t1}	ν_2	t_2	t_{t2}	P/s_3	P_4/s_3	H_{total}	Q_{e1}	Q_{e2}
20	437	70/100	0.006	471	21/100	0.006				11440 ³ 1.5x10 ⁵	
40	611	60/100	0.006	565	60/100	"				27440 ³ 1.8x10 ⁵	
60	714	15/24	0.018	848	15/24	"				26440 ³ 4.2x10 ⁵	
80	722	14/26	0.015	1130	14/26	"				26440 ³ 9.4x10 ⁵	

$$P_s = \frac{8 \cdot \nu \cdot D}{\mu}$$

$$\frac{\Sigma 1}{d_1} = 0.000625$$

$$\frac{\Sigma}{d_2} = 0.0005$$

$$2) P = 7 \text{ bar} = 700000 \text{ Pa} \quad A^* = 0,3 \text{ A}_2$$

$$T_0 = 400^\circ \text{K}$$

$$K = 1,4$$

$$R = 287,9$$

$$C_0 = \sqrt{1,4 \cdot 287,9 \cdot 400} = 401,5$$

$$\dot{m} \quad \dot{Q}_0 = \frac{P_0}{R \cdot T} = \frac{700000}{287,9 \cdot 400} = 6,07$$

$$\frac{T_0}{T_*} = \left(1 + \frac{K-1}{2} M_*^2 \right) \Rightarrow T_* = 333,3 \text{ K}$$

$$\frac{A_2}{A_*} = \frac{1}{M_2} \left(\frac{1 + \frac{K-1}{2} M_2^2}{\frac{K+1}{2}} \right)^{\frac{K+1}{2(K-1)}}$$

$$\approx 3,33 \quad \frac{A_2}{0,3 A_*} = \frac{1}{M_2} \left(\frac{1 + 0,2 M_2^2}{1,2} \right)^3 \Rightarrow M_2 = 2,78 \quad \checkmark$$

$$M_2 = 2,78$$

$$\frac{P_0}{P_*} = \left(1 + \frac{K-1}{2} M_*^2 \right)^{\frac{K}{K-1}}$$

$$\frac{P_0}{P_2} = \left(1 + \frac{K-1}{2} M_2^2 \right)^{\frac{K}{K-1}} = \frac{P_2}{P_1} = \left(\frac{1 + \frac{K-1}{2} M_*^2}{1 + \frac{K-1}{2} M_2^2} \right)^{\frac{K}{K-1}}$$

$$P_2 = 721,32 \text{ Pa}$$